

ZHAOTONG, China

WMO: 565860

Lat: 27.3517N

Lon: 103.7189E

Elev: 1950

StdP: 79.99

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.3	-3.3	-9.1	2.2	8.3	-7.5	2.5	6.1	7.4	12.3	6.4	12.3	1.5	0	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	28.7	17.4	27.5	17.4	26.4	17.1	19.7	25.1	19.3	24.5	18.9	23.8	3.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	16.8	21.6	17.9	16.3	21.2	17.4	15.9	20.9	66.5	25.3	64.6	24.8	63.0	24.0	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.5	4.8	DB	-6.0	30.9	1.5	1.0	-7.1	31.6	-7.9	32.1	-8.7	32.7	-9.8	33.4	(4)
			WB	-6.8	20.9	1.5	0.8	-7.9	21.4	-8.7	21.9	-9.6	22.3	-10.6	22.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	3.8	6.3	10.4	14.7	17.2	18.9	20.8	20.3	17.7	13.6	9.6	4.9
	DBStd	6.82	4.01	5.15	4.51	4.07	3.39	2.78	2.06	2.17	3.19	3.39	3.60	3.44
	HDD10.0	597	194	123	53	10	1	0	0	0	0	8	49	159
	HDD18.3	2132	451	337	248	119	62	26	4	8	50	151	261	415
	CDD10.0	1772	2	19	64	150	224	268	333	320	231	118	39	3
	CDD18.3	264	0	0	1	9	27	44	79	69	31	3	0	0
	CDH23.3	1379	0	2	26	115	203	218	340	302	156	18	0	0
	CDH26.7	195	0	0	2	13	38	38	45	39	20	0	0	0
Wind	WSAvg	1.7	1.7	2.1	2.3	2.2	1.9	1.6	1.5	1.4	1.5	1.5	1.5	1.5
Precipitation	PrecAvg	725	6	5	12	34	81	139	151	116	104	55	17	6
	PrecMax	895	20	20	35	75	280	249	341	254	179	158	68	29
	PrecMin	466	0	0	0	4	4	51	64	23	40	10	0	0
	PrecStd	103	5	5	10	20	56	57	59	51	36	31	16	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	23.0	26.1	28.4	29.8	29.7	29.4	29.3	28.8	25.3	22.2	18.6
		MCWB	7.5	8.8	10.4	13.2	15.1	18.1	19.0	18.6	17.8	15.8	11.4	8.0
	2%	DB	16.9	20.8	23.5	26.3	27.7	27.8	27.9	27.8	26.7	23.4	20.2	16.2
		MCWB	6.3	7.9	10.0	12.9	15.4	18.0	18.8	18.5	17.5	14.9	11.2	7.3
	5%	DB	14.5	18.6	21.6	24.5	25.7	25.9	26.7	26.4	25.1	21.5	18.3	13.6
		MCWB	5.9	7.2	9.5	12.5	15.3	17.7	18.6	18.4	17.1	14.3	10.6	6.8
	10%	DB	11.5	16.1	19.0	22.3	23.6	24.1	25.3	25.0	23.4	19.3	16.1	10.8
		MCWB	5.1	6.5	9.4	12.3	15.0	17.5	18.4	18.1	16.7	13.7	10.1	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	9.7	12.4	15.8	18.4	20.1	20.6	20.4	19.2	17.1	13.8	9.4
		MCDB	17.1	20.0	20.7	22.2	24.7	25.6	26.2	26.1	24.7	23.1	18.9	16.5
	2%	WB	7.1	8.6	11.3	14.7	17.4	19.4	19.8	19.6	18.6	16.1	12.2	8.0
		MCDB	15.4	18.2	19.9	21.9	23.8	24.7	25.1	25.0	24.0	21.5	18.2	14.4
	5%	WB	6.1	7.7	10.5	13.9	16.6	18.8	19.4	19.1	18.0	15.1	11.2	7.0
		MCDB	13.6	17.0	18.8	20.9	22.6	23.8	24.4	24.3	23.1	19.7	16.5	12.9
	10%	WB	5.0	6.9	9.7	13.0	15.8	18.2	18.9	18.6	17.4	14.1	10.3	5.9
		MCDB	11.5	15.4	17.3	20.0	21.3	22.6	23.8	23.4	22.1	18.1	15.0	10.6
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.7	11.5	11.1	9.7	7.8	7.9	8.2	8.0	7.6	9.3	8.9
		MCDBR	17.0	17.2	17.2	15.3	14.1	11.6	10.4	10.8	11.5	12.1	13.9	14.9
		MCWBR	8.0	6.9	6.1	4.6	4.4	3.9	3.2	3.4	3.6	4.4	5.9	7.4
	5% WB	MCDBR	15.7	15.8	14.7	12.9	11.3	9.0	8.6	9.1	9.7	10.3	12.0	13.5
		MCWBR	8.0	7.0	5.9	4.6	4.4	3.6	3.0	3.2	3.4	4.3	5.6	7.1
Clear-Sky Solar Irradiance	taub	0.367	0.383	0.496	0.511	0.470	0.455	0.456	0.477	0.461	0.463	0.419	0.385	
	taud	2.209	2.181	1.867	1.886	2.075	2.185	2.180	2.096	2.114	2.026	2.067	2.141	
	Ebn at Noon	865	883	800	801	832	839	837	819	821	789	797	822	
	Edh at Noon	123	137	200	201	167	149	149	161	154	158	141	127	
All-Sky Solar Radiation	RadAvg	2.82	3.77	4.39	4.95	4.75	4.08	4.67	4.37	3.62	2.83	2.96	2.48	
	RadStd	0.44	0.61	0.40	0.56	0.42	0.71	0.41	0.50	0.52	0.40	0.38	0.37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.46	N/A	N/A	+0.87	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	+0.46	N/A	N/A	+0.81	N/A	N/A	N/A	-92	+143	+54

Nomenclature: See separate page